MIL-M-45174B (PA)
31 July 1974
SUPERSEDING
MIL-M-45174A (ORD)
11 June 1962

MILITARY SPECIFICATION

MINE, ANTIPERSONNEL, M16A1 LOADING, ASSEMBLING AND PACKING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the loading, assembling and packing for one type of mine designated as Mine, Antipersonnel, M16A1.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116	- Preservation - Packaging, Methods of
MIL-P-223	- Powder, Black
MIL-A-2550	- Ammunition; General Specification for
*MIL-A-48078	- Ammunition, Standard Quality Assurance Provisions, General Specification for

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)

FSC: 1345

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MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

*(For meaning of asterisk (*) see 6.5)

DRAWINGS

U.S. ARMY PICATINNY ARSENAL

7548179 - Box, Wirebound Packing, Ammunition, for Mines, Antipersonnel, M16A1, or M16A2 with Fuze, Mine, Combination, M605

8796365 - Mine, Antipersonnel, M16A1 Assembly

8863609 - Box, Fiber, Packing, Ammunition, for Mines Antipersonnel, M16A1, or M16A2 with Fuze, Mine Combination, M605

(Copies of specifications, standards and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer).

3. REQUIREMENTS

3.1 Material. Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Mine Assembly. - The mine assemblies shall comply with all requirements specified on Drawing (dwg.) 8796365, and with all requirements specified in applicable specifications.

3.3 Moisture Content of Lead Azide. The moisture content of the lead azide at the time of loading shall not exceed 0.50.

3.4 Moisture Content of Black Powder. The moisture content of the black powder at the time of loading shall not exceed the percentage specified in Specification MIL-P-223.

3.5 Moisture Content of Normal Lead Styphnate. The moisture content of the normal lead styphnate at the time of loading shall not exceed 0.50 percent (%).

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3.6 Burning Time of Delay Assembly; Detonator Assembly (See 6.4). The burning time for the delay assembly - detonator assembly shall not be less than 22.5 milliseconds and shall not exceed 50.0 milliseconds.

3.7 Functioning. The mines shall function satisfactorily. Satisfactory functioning shall be defined as the bursting of the mine at a height of not less than one foot nor more than five feet above the surface of the ground, and with an average height of burst for all mines tested of not less than two nor more than four feet.

3.8 Workmanship. All parts shall be fabricated and finished (or fabricated and loaded) in a thorough, workmanlike manner, and all manufacturing, processing and assembly operations shall be correctly performed. The parts shall be clean and free of burrs, sharp edges, unblended radii, surface defects, chips, dirt, grease and oil (except where specifically required), corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part, nor shall the parts be contaminated by the cleaning agent. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined.

*3.9 First Article Inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

*4.1 Responsibility for Inspection. The responsibility for inspection shall comply with MIL-A-48078 (PA).

4.1.1 Submission of Product. At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted.

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.

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b. Drawing and specification number and date, together with identification and date of changes thereto.

c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.

d. Number of items in the lot.

e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

*4.2 Classification of Inspections. The following types of inspections shall be conducted on this item:

a. First article inspection

b. Quality conformance inspection

*4.3 First Article Inspection

4.3.1 Submission. Prior to the start of regular production, the contractor shall submit a first article sample (See 6.1b) as designated by the Contracting Officer for evaluation and testing in accordance with the provisions of 4.4.2.8 and 4.5.7. The first article sample shall consist of 50 complete mine assemblies which have been produced by the contractor or furnished by a supplier and which have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as will be used in regular production. Prior to submission,

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the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample, including a certificate of conformance for materials. A sample containing known defects will not be submitted unless specifically authorized by the Contracting Officer. A first article sample, or portion thereof, as directed by the Contracting Officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days or whenever a change occurs in manufacturing process, material used, drawing or specification such as to significantly affect product uniformity as determined by the Government.

4.3.2 Rejection. See MIL-A-48078 (PA).

4.4 Quality Conformance Inspection.

4.4.1 Inspection Lot Formation. See MIL-A-48078 (PA). Lot numbering, as required, shall be in accordance with MIL-STD-1168. In addition, inspection lots of Mine Assemblies shall contain:

a. Parts and parts assemblies of one interfix lot number from one manufacturer.

b. Fuzes of one interfix lot number from one manufacturer.

c. Tetryl of one interfix lot number from one manufacturer.

d. Lead Styphnate of one interfix lot number from one manufacturer.

e. Lead Azide of one interfix lot number from one manufacturer.

f. Black powder of one interfix lot number from one manufacturer.

g. TNT of one interfix lot number from one manufacturer.

*4.4.2 Examination. - See MIL-A-48078 (PA). The examinations listed in the following tables and subparagraphs shall be performed on inspection lots of the components or assemblies designated in the Classification of Defects and Test tables or paragraphs. Sampling plans for major or minor defects shall be in accordance with MIL-STD-105, Inspection Level II, unless otherwise specified. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.4.4.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH *	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.1	Detonator Assembly		8796342
			NEXT HIGHER ASSEMBLY 8796365

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Charge escaping		100%	3.2	Visual
2	Disc damaged (punctured or deep cut)		100%	3.2	Visual
3	Cup cracked or punctured		100%	3.2	Visual
Major A					
101	Charge missing		0.40%	3.2	Gage
Major B					
131	Length to flange, maximum (max.)		0.40%	3.2	Gage
132	Charge missing		0.40%	3.2	Visual
Minor					
201	Disc seal inadequate (disc separating from cup)		0.65%	3.2	Visual
202	Evidence of poor workmanship		1.00%	3.8	Visual

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CLASSIFICATION OF DEFECTS & TESTS

*PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.2	Propelling Charge Assembly				8796343
					NEXT HIGHER ASSEMBLY
					8796365
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined.				
Major B					
131	Color improper		0.40%	3.2	Visual
132	Bag damaged (hole, tear, weak spots or breaks)		0.40%	3.2	Visual
133	Marking misleading or unidentifiable		0.40%	3.2	Visual
134	Evidence of poor workmanship		0.40%	3.8	Visual
Minor					
201	Seam stitching inadequate (incomplete, opening, etc.)		0.65%	3.2	Visual

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH * 4.4.2.3	TITLE: Body Loading Assembly, Prior to Loading TNT, and Prior to Assembly of Detonator and Delay Assembly	SHEET 1 OF 1	DRAWING NUMBER 8796345 / NEXT HIGHER ASSEMBLY 8796345	
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major A					
101	Booster pellet missing		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.8	Visual

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH *	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.4	Body loading assembly, prior to loading TNT and prior to assembly of delay assembly	1	1	8796345
				NEXT HIGHER ASSEMBLY 8796345

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major A					
101	Detonator missing		0.40%	3.2	Visual
Major B					
131	Detonator cover disc missing or damaged		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.8	Visual

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH *	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.5	Body loading assembly, prior to loading TNT and prior to inserting in container assembly		8796345
			NEXT HIGHER ASSEMBLY 8796365

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major A					
101	Delay charge missing		0.40%	3.2	Visual
102	Propelling charge assembly missing or damaged		0.40%	3.2	Visual
103	Delay assembly missing		0.40%	3.2	Visual
104	Flash hole in delay washer obstructed		0.40%	3.2	Visual
105.	Delay washer missing		0.40%	3.2	Visual
Major B					
131	Delay assembly protrudes above face of delay holder		0.40%	3.2	Visual
132	Delay holder or detonator well protrude above surface of body		0.40%	3.2	Visual
Minor					
201	Delay washer crimp less than 360°		0.65%	3.2	Visual
202	Delay washer inverted		0.65%	3.2	Visual
203	Flash disc of flash pipe assembly missing or damaged (punctured or with a deep cut)		0.65%	3.2	Visual
204	Evidence of poor workmanship		0.65%	3.8	Visual

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH *	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.6	Body loading assembly, after loading TNT, less filler hole plug, and prior to inserting in container assembly.		8796345
			NEXT HIGHER ASSEMBLY 8796365

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined.				
Major A					
101	TNT missing		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

*PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.7	Body loading assembly, prior to inserting into container assembly				8796345
					NEXT HIGHER ASSEMBLY 8796365
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined.				
Major B					
131	Filler hole plug missing or damaged (punctured or deep cut)		0.40%	3.2	Visual
132	Explosive visible on any outer surface or through any opening		0.40%	3.2	Visual
Minor					
201	Seal missing or inadequate at filler hole plug or propelling charge housing		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.8	Visual

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH *	TITLE	SHEET	DRAWING NUMBER
4.4.2.8	Body loading assembly	1 OF 1	8796345
			NEXT HIGHER ASSEMBLY 8796365

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major B					
131	Shipping plug missing or loose		0.40%	3.2	Visual/Manual
132	Top of container inadequately secured to its side		0.40%	3.2	Visual
133	Container assembly damaged (punctured or dented)		0.40%	3.2	Visual
134	Shipping gasket or bushing washer missing		0.40%	3.2	Visual
Minor					
201	Marking misleading or unidentifiable, including incorrect color code		0.65%	3.2	Visual
202	Protective coating damaged, other than slight scratches		0.65%	3.2	Visual
203	Burr, projection, raised metal, or sharp edge		0.65%	3.2	Visual
204	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 8863609
*4.4.2.9	Box, fiberboard, packing, unsealed				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:	None defined				
Major A: 101.	Mine, spool, wrench or container of fuzes missing		0.40%	3.2	Visual
Minor: 201.	Evidence of poor workmanship		0.65%	3.8	Visual

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
*4.4.2.10	Sealed barrier bag (prior to packing in wire-bound box)		8863609
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:	None defined				
Major B:					
131.	Waterproofness of barrier bag destroyed by rupture, puncture, or separation of heat seal		0.40%	3.2	Visual
132.	Evidence of crushed carton (see NOte 1)		0.40%	3.2	Visual
Minor:					
201.	Heat seal of barrier bag located so that one or more subsequent seals cannot be applied		0.65%	3.2	Visual
202.	Marking of bag misleading or unidentifiable		0.65%	3.2	Visual
203.	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES:

NOTE 1. - Package should be opened for inspection if there is evidence of crushing.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 7548179
*4.4.2.11	Box, packing, wirebound (prior to sealing)				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:	None defined				
Major B:					
131.	Barrier bag missinggor damaged		0.40%	3.2	Visual
132.	Packing component (filler) missing or improperly positioned		0.40%	3.2	Visual
Minor:					
201.	Evidence of poor workmanship		0.65%	3.8	Visual

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 7548179
*4.4.2.12	Box, packing, wirebound (sealed)				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:	None defined				
Major B:					
131.	Box damaged to the extent that contents are exposed or liable to become exposed		0.40%	3.2	Visual
132.	Wirebinding missing, broken or loose		0.40%	3.2	Visual
Minor:					
201.	Wirebinding mislocated or improperly engaged		0.65%	3.2	Visual
202.	Contents loose		0.65%	3.2	Visual
203.	Marking misleading or unidentifiable		0.65%	3.2	Visual
204.	Metallic seal missing, unsealed or incorrectly positioned		0.65%	3.2	Visual
205.	Cleats or battens split or broken		0.65%	3.2	Visual
206.	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES:

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4.4.3 Testing

4.4.3.1 Moisture Content of Lead Azide (See 3.3) Major Defect A. The moisture content of the lead azide shall be determined at the time of loading not less than once for each eight hours of production, or portion thereof. The sample shall be selected just prior to loading and the moisture content determined as specified in 4.5.1. Composite samples shall not be used in this determination. If the moisture content exceeds the requirement specified, the lead azide shall be rejected. If this determination is made after loading, has begun, components loaded therewith shall be rejected.

4.4.3.2 Moisture Content of Black Powder. (See 3.4) Major Defect A. The moisture content of the black powder shall be determined at the time of loading not less than once for each eight hours of production, or portion thereof. Two grams of black powder shall be selected just prior to loading and the moisture content determined as specified in 4.5.2. Composite samples shall not be used in this determination. If the moisture content exceeds the requirement specified, the black powder shall be rejected. If this determination is made after loading has begun, components loaded therewith shall be rejected.

4.4.3.3 Moisture Content of Normal Lead Styphnate (See 3.5) Major Defect A. The moisture content of the normal lead styphnate shall be determined at the time of loading not less than once for each eight hours of production, or portion thereof. The sample shall be selected just prior to loading and the moisture content determined as specified in 4.5.3. Composite samples shall not be used in this determination. If the moisture content exceeds the requirement specified, the normal lead styphnate shall be rejected. If this determination is made after loading has begun, components loaded therewith shall be rejected.

4.4.3.4 Binder-Lubricant Content of Tetryl (See dwgs. 8796339 and 8796340) Major Defect A. The binder lubricant content of the tetryl shall be determined for

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each lot of tetryl at the time of loading. A sample of five grams shall be selected just prior to loading and the determination made as specified in 4.4.4. Composite samples shall not be used in this determination. If the binder-lubricant content exceeds the requirement specified, the tetryl shall be rejected. If this determination is made after loading has begun, components loaded therewith shall be rejected.

4.4.3.5 Burning Time of Delay Assembly - Detonator Assembly (See 3.6) Major Defect A. - Eighty (80) delay assembly-detonator assemblies shall be selected from each lot for test. The test shall be performed as specified in 4.5.6. If two or more delay assembly-detonator assemblies fail to comply with the applicable requirements, the lot shall be rejected. Any delay assembly-detonator assembly whose delay time is lost due to equipment failure or error in procedure, shall be replaced in order to determine if the lot complies with the delay time requirements.

4.4.3.6 Functioning (See 3.7) - Major Defect A.

4.4.3.6.1 Beginning with the first lot produced and continuing until three (3) consective lots have complied with the acceptance criteria specified, a sample of eighty (80) assemblies shall be selected from each lot for this test. The test shall be performed as specified in 4.5.7. The lot shall be rejected if:

a. Three (3) or more mines burst at a height of less than one (1) foot or more than five (5) feet.

b. The average height of burst is less than two (2) feet or more than four (4) feet.

c. Two (2) mines fail to burst.

4.4.3.6.2 After three (3) consecutive lots have met the criteria of 4.4.3.4.1, a sample of fifty (50) assemblies shall be selected from each lot for this test. The lot shall be rejected if:

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a. Two (2) or more mines burst at a height of less than one (1) foot or more than five (5) feet.

b. The average height of burst is less than two (2) feet or more than four (4) feet.

c. Two (2) mines fail to burst.

4.4.3.7 Heat sealed seam test (see dwg. 8863609) Major Defect B. - Ten (10) fiberboard boxes in sealed barrier bags shall be selected from each lot for this test. If two or more packages show evidence of heat sealed seam separation, the lot shall be rejected.

*4.4.4 Inspection equipment. - The inspection equipment required to perform the examinations and tests prescribed herein is described in the 'Paragraph Reference/Inspection Method' column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078.

*4.5 Test Methods and Procedures

*4.5.1 Materials, Components and Processes. Compliance with all requirements of Section 3 of this specification shall be ascertained by current and continuing examination of inspection and test data to determine that all components (parts, subassemblies and materials) have been inspected and tested and found to comply with their respective drawings and specification requirements, and that all specified manufacturing processes have been followed.

*4.5.1.1 Visual and Mechanical Inspections. The visual, manual and mechanical inspections and tests shall verify compliance with the requirements of Sections 3 and 5 of this specification in accordance with 4.4.2 herein.

4.5.2 Moisture Content of Lead Azide. A tared dried aluminum dish, 2 inches in diameter, shall be covered with a sheet of polytetrofluoroethylene, 2 1/2 inches square, and sufficiently flat to form a tight cover. An accurately weighed portion of approximately two grams of the sample shall be transferred to a moisture dish and the contents and the dish shall be transferred to an oven and dried at 95 degrees Centigrade (°C) plus or minus (±) 2°C, for 2 hours. The cover shall be placed in a sulfuric acid desiccator during the drying period. After drying, the dish shall be quickly transferred to the desiccator and allowed to cool to room temperature. The polytetrofluoroethylene cover shall be

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placed on the dish and the assembly weighed. The loss in weight shall be calculated to percentage of moisture content in the lead azide to determine compliance with 3.3 as follows:

$$\text{Percentage of moisture} = \frac{(A - B) \times 100}{A}$$

Where:

A = the weight of the sample before drying

B = the weight of the sample after drying

4.5.3 Moisture Content of Black Powder. The moisture content of the black powder shall be determined as specified in Specification MIL-P-223.

4.5.4 Moisture Content of Normal Lead Styphnate. The moisture content of the normal lead styphnate shall be determined as follows:

Transfer, quickly, approximately two grams of normal lead styphnate to an uncovered tared pyrex dish, approximately 55mm in diameter and 30mm in height. Weigh the sample as quickly as possible to 0.0001 gram and place in an oven maintained at 70° ± 2°C and dry to constant weight. At the end of the drying period, remove the sample from the oven and immediately place it in a desiccator containing anhydrous calcium chloride. Allow the sample to cool in the desiccator for approximately 20 minutes and then, as quickly as possible, weigh to the nearest 0.0001 gram. Calculate the percentage of moisture in the sample from the loss in weight.

4.5.5 Binder-Lubricant Control of Tetryl. An accurately weighed portion of five grams shall be transferred to a tared, medium porosity, sintered, glass filtering crucible. The specimen shall be extracted with four 10-milliliter (ml) portions of acetone, maintaining each portion in contact with the residue for approximately five minutes before applying suction. Any lumps which may persist shall be broken up and the sides of the crucible washed down with each 10-ml portion of acetone to insure complete removal of the tetryl.

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The residue shall be aspirated until the odor of acetone is no longer detectable. The crucible and contents shall be placed in a drying oven maintained at 100° to 105°C for 1/2 hour. cooled in a desiccator and weighed. The above extracting shall be repeated, drying and weighing with individual 10-ml portions of acetone until the loss in weight on subsequent extractions totals less than 0.005 gram. The weight of residue in the crucible shall be calculated to the percentage of total binder-lubricant in the specimen as follows:

$$\text{Total percent of binder-lubricant} = \frac{100(A)}{B}$$

Where:

A = weight (in grams) of residue in crucible
B = weight (in grams) of original specimen of tetryl

4.5.6 Buring Time of Delay Assembly - Detonator Assembly. The delay assembly-detonator assembly shall be tested for burning time using methods and equipment in accordance with 4.3.4 (See 6.4).

4.5.7 Functioning. The mines, less fuzes, shall be immersed in water to a depth of not less than six inches above the top of the mines, for not less than twenty-four nor more than thirty hours. The water shall contain a wetting agent such as naccatol. After removal of the mines from the water, assemble with M605 combination mine fuzes. The assemblies shall be placed in a hole in the ground to such a depth that the top of the mine will be covered, but the release spring of the fuze uncovered when the hole is refilled and the ground made level. The hole shall then be filled with firmly packed sand, fine gravel or dirt. The fuze shall be functioned and observation made for satisfactory functioning to determine compliance with 3.7.

4.5.8 Heat sealed seam test. - This test shall be performed as specified in MIL-P-116.

5. PREPARATION FOR DELIVERY

5.1 Packing

5.1.1 Level A. Packing for Mine, Apers, M16A1, shall be in accordance with dwgs. 7548179 and 8863609.

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5.2 Marking. Marking shall be as specified on dwg. 7548179.

6. NOTES

6.1 Ordering Data. Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Provisions for submission of first article samples.
v. Provisions of MIL-A-48078.

6.2 Submission of designs for approval. - (See 6.2.3 of MIL-A-48078 (PA). Designs shall be submitted for approval to Commander, Picatinny Arsenal, ATTN: SARPA-QA-T-T, Dover, New Jersey 07801.

6.3 Distribution of ammunition data cards. - Distribution of data cards shall include the following: Commander, Picatinny Arsenal, ATTN: SARPA-QA-A-M, Dover, New Jersey, 07801

6.4 If the loading plant is responsible for the complete mine, that is, loading of the detonator and delay assemblies and the mine, requirements relating to burning time of delay assemblies may be omitted and testing shall be conducted in accordance with requirements of 3.7.

6.5 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian:
Army-PA

Preparing Activity:
Army-PA

Project Number: 1345-A-078

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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☐ YES ☐ NO *(If "Yes", in what way?)*

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